

1. Here is a simple game. Start with a pile of raisins. A turn consists of a player taking 1, 2, 3 or 4 raisins. The last player to have a turn (that is, the player who takes the last raisin) wins. Can you find an optimal strategy? Who should win – the player who goes first or the player who goes second?
2. Tweak the previous game to allow each player to take 1 through n raisins. How has the game changed?
3. Start with two piles of raisins. A player's turn consists of taking *either* some (any number of) raisins from a single pile *or* the same number of raisins from both piles. The person to take the last raisin wins. Can you find an optimal strategy?